IEOR E4101 Probability, Statistics and Simulation **Fall 2025**

Homework 5

Instructor: Henry Lam *Due: October 27, 2025*

Problem 1 College-Pro Painting does home interior and exterior painting. The company uses inexperienced painters that do not always do a high-quality job. It believes that its painting process can be described by a Poisson distribution with an average of 4.8 defects per 400 square feet of painting. What is the probability that a 400-square-foot painted section will have fewer than 6 blemishes? What is the probability that at least two out of six randomly sampled (non-overlapping) sections of size 400 square feet will have fewer than 6 blemishes?

Problem 2 Approximately 80,000 marriage took place in the state of New York last year. Estimate the probability that for at least one of these couples

(a) both partners were born on April 30;

(b) both partners celebrated their birthday on the same day of the year.

State your assumptions.

Problem 3 In class, we saw that exponential distribution is memoryless. Similarly, is geometric distribution memoryless? To be precise, let $X \sim \text{Geo}(p)$ for some p in (0,1); that is, X is a geometric random variable with the parameter denoting probability of success $= p$. Then, for all positive integers m and n, is

$$P(X > m + n \mid X > m) = P(X > n)?$$

Problem 4 A contractor purchases a shipment of 100 transistors. It is his policy to test 10 of these transistors and to keep the shipment only if at least 9 of the 10 are in working condition. If the shipment contains 20 defective transistors, what is the probability it will be kept?

Problem 5 You arrive at a bus stop at 10 o'clock, knowing that the bus will arrive at some time uniformly distributed between 10 and 10:30. What is the probability that you will have to wait longer than 10 minutes? If at 10:15 the bus has not yet arrived, what is the probability that you will have to wait at least an additional 10 minutes?

Problem 6 A serial system has 10 components, each of which has a lifetime that is exponentially distributed, independently from each other. Their expected lifetimes are $1.1, 1.2, 1.3, \ldots, 2.0$ hours respectively. The system functions only if all the components are functioning. Suppose that at time 0, the system is functional. What is the probability distribution of the lifetime of the system? What are the expectation and standard deviation of this lifetime?

Problem 7 Romeo and Juliet have a date at a given time, and each, independently, will be late by an amount of time that is exponentially distributed with parameter λ. What is the probability density function of the difference (just the difference, not the absolute difference, so it can be negative if Romeo comes early) between Romeo's and Juliet's arrival time?